Identifying Energy Poverty Risk at Small Area Level in Ireland: A Data Mining Approach

Source code and datasets link at: Group-38-Energy-Poverty

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***Abstract*—Identifying energy-poor communities at actionable spatial resolution remains an open challenge in Ireland, where 28% of households are affected but no Small Area level classifier exists. This study takes a different approach: we integrate three publicly available government datasets, the CSO Small Area Population Statistics, the SEAI Building Energy Rating database, and CSO boundary files, and train a supervised binary classifier across all Irish Small Areas. Following the CRISP-DM process, we engineer features from BER physical measurements, Census socio-economic counts, income proxy rates, and a spatial lag derived from Queen contiguity weights. A tuned Random Forest achieves a test accuracy of 77.40%, balanced accuracy of 75.31%, and AUC-ROC of 0.8393, with cross-validation confirming the result is stable and not an artefact of the train-test split. The model identifies over four thousand Small Areas as High Risk, just over one fifth of the national total. CO_2 rating and wall U-value dominate the predictive signal, with housing age and spatial neighbourhood context adding meaningful lift. The output is a national risk map that SEAI programme managers could use directly to prioritise Warmer Homes Scheme outreach.**

***Index Terms*—energy poverty, BER, random forest, CRISP-DM, Irish small areas, data mining, feature engineering**

I. Introduction

Energy poverty — the inability to maintain adequate warmth and basic energy services at home — is not simply a billing problem [3]. In Ireland, it intersects decades of underinvestment in housing quality, persistently low incomes (particularly in rural areas), and an energy market that has experienced significant volatility since 2021. Wholesale gas prices more than tripled between 2021 and late 2022, and by 2023 household electricity costs had risen by over 40% [7]. The government's own National Energy Poverty Action Plan 2022–2030 acknowledges that housing condition, income level, and fuel price exert simultaneous pressure on the same household, and that their combined effect differs sharply by region and tenure type [8]. Programmes such as the Warmer Homes Scheme and the Better Energy Communities initiative rely on households coming forward for support. This creates a structural gap: the areas with the poorest housing stock often exhibit the lowest application rates — not because the problem is smaller, but because access barriers are higher.

This is where a spatially granular risk classifier becomes particularly valuable. Ireland's Small Areas — the finest Census geography, covering roughly 65–80 households each on average — offer a resolution at which meaningful targeting is feasible. The SEAI BER database contains over one million property-level energy assessments. When merged with the 2022 Census Small Area Population Statistics (SAPS) and mapped to Small Area boundaries, the resulting dataset captures both the physical condition of the housing stock and the socio-economic profile of its residents.

This study trains a binary classifier on that merged dataset. The questions we set out to answer were:

RQ1 Can BER and Census features, combined at Small Area level, support a machine learning classifier that reliably identifies high energy poverty risk?

RQ2 Which feature group drives model performance most — physical building characteristics or socio-economic factors?

RQ3 How does data cleaning and imputation affect the completeness and usability of the final modelling dataset?

Section II surveys the relevant literature; Section III details the CRISP-DM pipeline; Section IV covers model evaluation and results; Section V closes with limitations and next steps.

II. Related Work

A. Defining and Measuring Energy Poverty

There is no single agreed definition of energy poverty, and the choice of definition affects who gets counted. One of the earliest and most widely used benchmarks holds that a household spending more than one tenth of its income on keeping warm qualifies as fuel poor, a threshold that has shaped policy thinking across Europe for decades. The rule is intuitive, but Hills [10] later pointed out that it treats a high-income household spending 11% the same as a low-income household spending 11%, which is clearly not the same situation. His Low Income High Costs (LIHC) indicator, now standard in UK policy, tries to fix this by requiring both conditions to hold simultaneously. In Ireland, SEAI adapted the same underlying logic to local energy prices and stock characteristics [7].

The measurement problem is even sharper in rural Ireland. Scott et al. [16] ran the European Energy Poverty Index against

Irish SILC data and found that income-based thresholds consistently miss rural households whose main fuel is heating oil, a fuel that does not show up clearly in standard energy expenditure surveys and whose price swings more violently than electricity or gas. Their finding is one reason we anchor our target variable to BER rating rather than self-reported spending: at Small Area level, expenditure data simply is not available, and a physical proxy grounded in assessed building performance is both more practical and arguably more stable.

B. Building Energy Ratings as Poverty Proxies

The most directly relevant prior work for Ireland is Kelly et al. [12], who built a Household Energy Poverty Risk Index (HH-EPRI) by combining BER score, income quintile, and dwelling size for approximately 18,641 Small Areas using Census 2016 data. They showed that low-rated properties cluster strongly in rural western counties, and that when poor BER performance coincides with low income and large floor area the risk multiplies. Our work uses their framing as a starting point, but moves from a composite index to a trained classifier: rather than summing weighted scores, we let a Random Forest learn the relationship from data. The practical difference is that a classifier can be retrained as new BER records are published, without requiring researchers to manually re-calibrate index weights.

Two European studies reinforce choices we made in feature engineering. Charlier and Legendre [11] studied French thermal renovation records and found that building age predicted energy inefficiency more reliably than almost any other variable, outperforming floor area, tenure type, and heating system. Ireland's situation is comparable: the country's pre-1981 stock was built largely before Part L thermal regulations existed, and it shows in the BER data. That is why we include `old_housing_rate` as a feature rather than treating vintage as background context. The same logic applies to data integration: no single source tells the full story, which is why we bring together three government databases rather than relying on BER records alone.

C. Machine Learning for Socio-Spatial Classification

Random Forest has a reasonably strong track record in energy certificate prediction. Ali et al. [2] tested ensemble-based machine learning models, including Random Forest and XGBoost, against traditional approaches on Irish residential housing data and found that tree-based ensemble models consistently outperformed Logistic Regression for predicting building energy performance. Tree-based models, in particular, better capture non-linear interactions between building age and construction type. Part of the reason is that the relationship between building features and energy rating is not linear — for example, older properties with cavity walls behave differently from older properties with solid stone walls, even if their floor areas are similar. Tree-based models handle these interactions automatically without the need to specify them in advance.

Class imbalance is a recurring problem in this space. When only 20-odd percent of areas are high-risk, a naive classifier can achieve 78% accuracy by predicting "not at risk" for everyone, which is useless for targeting purposes. Zheng and McKenna [20] found that class weighting outperformed resampling techniques for energy poverty identification on UK administrative data with a 6:1 class imbalance, a finding consistent with our experience with the 4:1 split in our dataset.

The spatial lag point deserves attention because it is sometimes treated as a minor refinement. Pokhriyal and Jacques [14] showed a 2–4% accuracy improvement from including neighbourhood deprivation context across African sub-national units, which is non-trivial when your baseline is already 75%. Our Moran's I result ($I = 0.276$, $p = 0.001$) confirms that Irish Small Area risk scores are not spatially random, so leaving out neighbour information would mean ignoring a real signal in the data [13].

D. CRISP-DM in Energy Data Mining

CRISP-DM [4] is the process framework we used to structure the study. Its value is not in being a checklist, it is in forcing the business understanding phase to happen before any data is touched. In energy poverty work, that matters: the choice of target variable, the decision about what counts as "high risk", and the selection of proxy features all need to be grounded in policy context before a model is trained, otherwise the evaluation metrics become hard to interpret. Fayyad et al. [9] made essentially the same point in the original KDD process paper, arguing that domain knowledge should constrain feature engineering rather than being bolted on afterwards.

Wirth and Hipp [19] note in the canonical CRISP-DM reference that data preparation typically absorbs 60–80% of total project time. We found this accurate: standardising the `SA_Code` field across the BER database, which stores the same identifier as floats, variable-length strings, and slash-delimited multi-area codes depending on the record, consumed more calendar time than any modelling step.

III. Data Mining Methodology

This study follows the CRISP-DM process model [19] across six phases: (1) *Business Understanding*, establishing energy poverty identification as a binary spatial classification problem; (2) *Data Understanding*, exploratory profiling of 18,919 Small Areas, 1.35 million BER records, and 200+ Census variables; (3) *Data Preparation*: multi-source integration, feature engineering, and MissForest imputation; (4) *Modelling*: training and tuning a Random Forest classifier; (5) *Evaluation*: cross-validation, AUC analysis, and geographic validation; (6) *Deployment*: production of a national energy poverty risk map.

A. Data Sources

Three primary data sources were integrated:

1) **CSO Small Area Population Statistics (SAPS) 2022**, 18,919 Small Areas, 200+ variables covering demographics, employment, housing, and transport [5].

2) **SEAI BER Public Database**, 1,346,555 property-level energy assessments, including `EnergyRating`, CO_2 rating, ground floor area, wall U-value, delivered energy for main space heating, and total delivered energy [17].
3) **CSO 2022 Small Area boundary GeoJSON**, spatial polygon geometries enabling choropleth mapping and spatial weight computation [6].

B. Data Integration and Cleaning

The BER database needed a lot of cleaning up before we could use it. We started with 1,346,555 records but 327,725 of them were missing a `SA_Code` (Small Area Code) so we had to get rid of them. The remaining records had values that were not formatted consistently. They were sometimes numbers, sometimes strings and sometimes they had multiple areas like "267103001/01". To fix this we did a three-step cleaning process. We made all the codes into 9-digit strings with zeros at the start. We split the multi-area entries at the "/" and kept the first valid code. This step helped us some records that would have been lost so we ended up with 1,018,829 property records with just one record that we could not fix.

We then took the cleaned up BER data, put it together at the Small Area level. For things like `EnergyRating` we used the common value and for numbers we took the average. This gave BER coverage for 16,827 Small Areas (88.9% of 18,919 total). The remaining areas, which is over 11% were filled in using a method called MissForest [18], which is good for dealing with different types of data later after complete integration of datasets. After we put the data together we combined the BER data with the Small Area file and the 2022 SAPS data using the `SA_PUB2022` identifier.

From the Census data we made a set of indicators that show the economic conditions of the areas. These included things, like how many people are unemployed, how many old people live there, how many households do not have a car, how many people rent privately, how many households do not have central heating and how many old houses are there. We also made some measures to try to capture how rich or poor the areas are, like how many people have medical cards how many people rely on welfare how many old people get payments how many people live in social housing and how many families have children. We made sure all the variables were proportions so we could compare areas of sizes.

C. Exploratory Data Analysis

Fig. 1 shows a broadly bell-shaped distribution centred at C1–C2, with very few areas at the highest efficiency band and negligible representation at F and G. Approximately 37.3% of areas fall in the D1-or-below band, consistent with Ireland's large stock of pre-1981 housing built before modern insulation requirements [7]. Fig. 2 shows 69.7% urban and 30.3% rural Small Areas. While this reflects Ireland's Census 2022 population split, rural areas are disproportionately affected due to older housing and heavy reliance on oil heating. Fig. 3 establishes the 11.1% BER coverage gap, concentrated in remote rural Small Areas where infrequent property

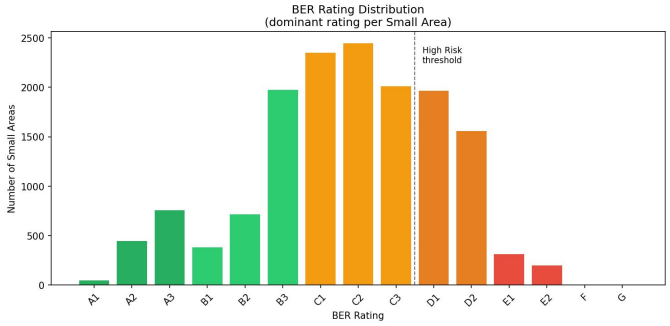

Fig. 1. BER rating distribution by dominant rating per Small Area. The dashed line marks the High Risk threshold (BER D2 or worse). The distribution centres at C1–C2; 4,026 areas (21.3%) fall at or below the threshold.

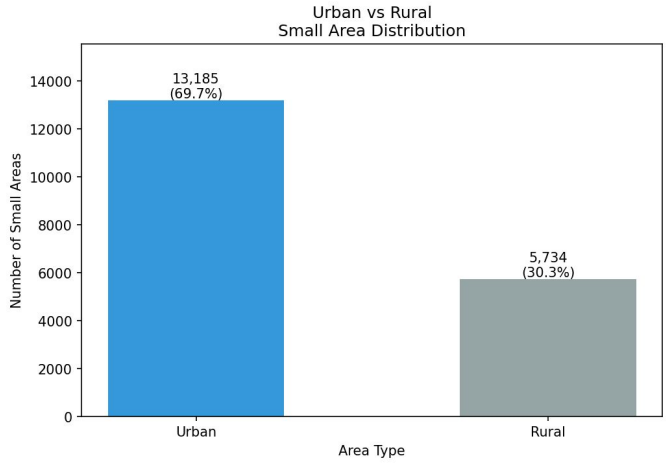

Fig. 2. Urban vs. rural Small Area distribution ($n = 18{,}919$). Urban areas (69.7%, $n = 13{,}185$) substantially outnumber rural areas (30.3%, $n = 5{,}734$), motivating an urban bias correction applied to `old_housing_rate`.

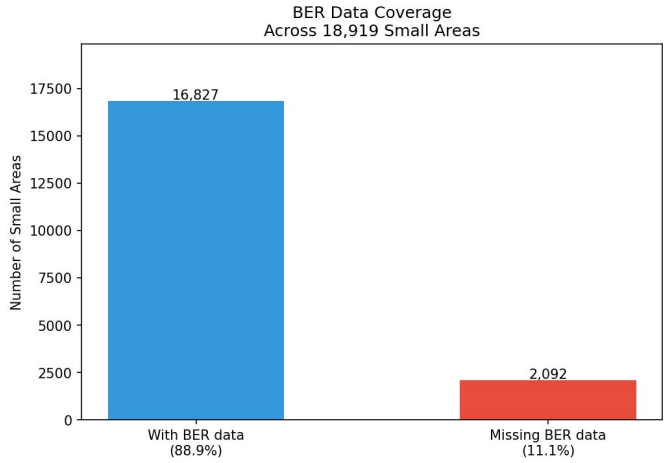

Fig. 3. BER data coverage across 18,919 Small Areas. Of these, 16,827 (88.9%) have valid BER assessments; 2,092 (11.1%) required MissForest imputation.

transactions mean no assessment has ever been triggered. Since these uncovered areas overlap significantly with the most structurally vulnerable housing, simple deletion was not appropriate and directly motivated the MissForest imputation strategy described in Section III-F.

D. Feature Engineering

Nineteen features were assembled across five categories. Six BER physical features (`CO2Rating`, `UValueWall`, `GroundFloorArea`, `DeliveredEnergyMainSpace`, `TotalDeliveredEnergy`, `property_count`) were aggregated from the SEAI BER database using modal rating and mean numeric values per Small Area. Five socioeconomic rates were computed from SAPS Census 2022 counts — unemployment_rate (total unemployed / total population), `old_housing_rate` (pre-1980 dwellings across five vintage bands / total dwellings), `elderly_rate` (population 65+ / total population), `private_renter_rate` (private rented households / total households), and `no_central_heating_rate` (households without central heating / total households).

Five income proxy rates — `medical_card_rate`, `old_age_payment_rate`, `social_housing_rate`, `welfare_rate`, and `family_with_children_rate` — were computed from SAPS T2, T4, T6, and T14 tables. `T5_1T_H` and `SA_URBAN_AREA_FLAG` were carried directly from the merged files. Global Moran's I computed on the composite risk score confirmed statistically significant spatial autocorrelation (I = 0.276, p = 0.001), justifying inclusion of `risk_score_spatial_lag` - the row-standardised mean risk score of each area's Queen contiguity neighbours [13].

E. Target Variable Construction

The binary target `energy_poverty_risk` was built by mapping each Small Area's dominant BER rating onto a 1–15 ordinal scale (A1=1 through G=15). Areas scoring ≥ 10 (D1 or worse) were labelled *High Risk*, consistent with the SEAI EEOS energy poverty threshold [16]. Before constructing the composite risk score, `old_housing_rate` was capped at 0.45 for urban areas to account for the fact that older urban buildings are more likely to have seen some level of retrofit compared to rural equivalents. This gave a final distribution of 4,026 High Risk (21.3%) and 14,893 Not At Risk (78.7%) Small Areas. The 4 :1 class imbalance was handled through `class_weight='balanced'` during training, which penalises minority class misclassification by a factor of 3.7. We explicitly ruled out SMOTE — Irish Small Areas are fixed geographic units and synthetically interpolating between them would generate areas that do not exist and cannot be targeted for any real intervention.

F. MissForest Imputation

Of the 18,919 Small Areas, 2,153 had no BER data across five columns – `CO2Rating`, `UValueWall`, `GroundFloorArea`, `DeliveredEnergyMainSpace`, and `TotalDeliveredEnergy`. Rather than dropping these areas or substituting column means, we used MissForest [18], which trains a Random Forest on the observed portion of the data and repeatedly predicts the missing entries until the estimates stabilise. Seven Census features drove the imputation: `unemployment_rate`, `elderly_rate`, `old_housing_rate`, `private_renter_rate`, `no_central_heating_rate`, `T5_1T_H` and `SA_URBAN_AREA_FLAG`. Mean imputation was not an option here - BER variables do not behave linearly with respect to these Census predictors, and a naive average would have consistently underestimated risk in the remote rural areas where the coverage gap is largest.

G. Model Training and Hyperparameter Optimisation

Four classifiers were evaluated — Logistic Regression, Random Forest, Gradient Boosting, and XGBoost — each wrapped in an `sklearn` Pipeline with `SimpleImputer` to prevent data leakage across the 5-fold cross-validation folds. Random Forest came out consistently ahead on AUC-ROC and was taken forward for tuning.

Optuna [1] with a TPE sampler was run over 100 trials optimising 5-fold stratified CV accuracy across the key Random Forest hyperparameters. The best trial returned `n_estimators=278`, `max_depth=17`, `min_samples_split=3`, `min_samples_leaf=5`, and `max_features=0.5` with a CV accuracy of 81.65%. Optuna returned `class_weight=None` which was overridden to `balanced` — without this the minority High Risk class is consistently under-predicted given the 4:1 class imbalance.

A manual depth sweep then tested depths 8, 10, 12, 15, and 17 using the remaining best parameters. Depth=12 achieved 77.40% test accuracy with an overfitting gap of 8.73%, while depth=17 recovered only 1.85% more accuracy at the cost of 4.54% additional overfitting (gap=13.2%). Depth=12 was therefore selected as the final configuration, giving the better bias-variance trade-off for generalisation to unseen areas.

The final model was trained on an 80/20 stratified train-test split of 18,916 Small Areas with the locked parameters (see Table I) and validated using 5-fold stratified cross-validation to confirm generalisation across the full dataset.

IV. Evaluation and Results

A. Model Performance

Table I summarises the performance of the final tuned Random Forest classifier. The test accuracy of the model is 77.405-fold CV accuracy of $77.58\% \pm 0.55\%$, balanced accuracy of 75.31%, and AUC-ROC of 0.8393. The small difference between CV and test accuracy The fact that the model generalises to unseen data without (0.18) confirms this. fitting to this training fold.

The trade-off between precision and recall of the High Risk class. precision=0.48, recall=0.72 is the result of a design decision: New `class_weight='balanced'` was

TABLE I
FINAL RF MODEL PERFORMANCE ($n = 18{,}916$ SMALL AREAS)

Metric	Value	Notes
Train Accuracy	86.13%	`max_depth=12`
Test Accuracy	77.40%	20% holdout
CV Accuracy (5-fold)	$77.58\% \pm 0.55\%$	Stratified
Overfitting Gap	8.73%	Train − Test
Balanced Accuracy	75.31%	Class-weighted
AUC-ROC	0.8393	Binary classifier
Precision (High Risk)	0.48	Recall–precision trade-off
Recall (High Risk)	0.72	`class_weight=balanced`
F1-Score (High Risk)	0.57	Imbalanced target (4:1)

used to favour recall, limiting the chances of overlooking truly vulnerable communities, at the cost of some false positive rate. In a policy context this is preferable: a false positive wastes limited on retrofit resources, but a false negative accompanies denial to a vulnerable household.

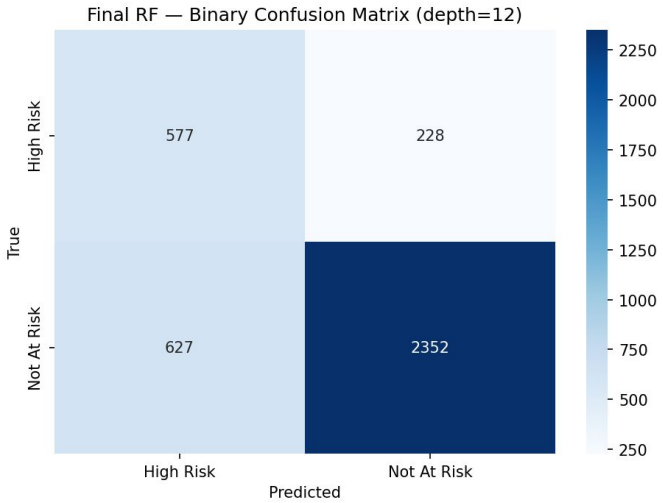

Fig. 4. Binary confusion matrix for the final Random Forest (`max_depth=12`). Of 805 High Risk test areas, 577 are correctly identified (recall = 0.72). Of 2,979 Not At Risk areas, 2,352 are correctly classified.

Table II shows how the trade-off between bias and variance is biased. guided final model choice. Depth=8 produced the smallest overfitting gap (2.85%) but the lowest test accuracy (73.89%). Depth=17, the Recommended Settings (Optuna), only a slight improvement over depth=12. (79.25% vs. 77.40%) while more than doubling the overfitting gap (13.27% vs. 8.73%). Depth=12 provides the optimal balance between stability of predictive performance and generalisation.

B. Feature Importance

Fig. 5 shows that CO_2Rating dominates with an importance of approximately 0.25, nearly double that of the second-ranked feature UValueWall (≈ 0.13) [12]. CO_2 emissions intensity encodes both the thermal quality of the building envelope and

TABLE II
DEPTH SWEEP, BIAS-VARIANCE TRADE-OFF ANALYSIS

Depth	Train	Test	CV	Gap
8	76.74%	73.89%	74.29%	2.85%
10	81.82%	75.66%	75.92%	6.16%
12	**86.13%**	**77.40%**	**77.58%**	**8.73%**
15	90.83%	78.83%	78.96%	12.00%
17	92.52%	79.25%	79.78%	13.27%

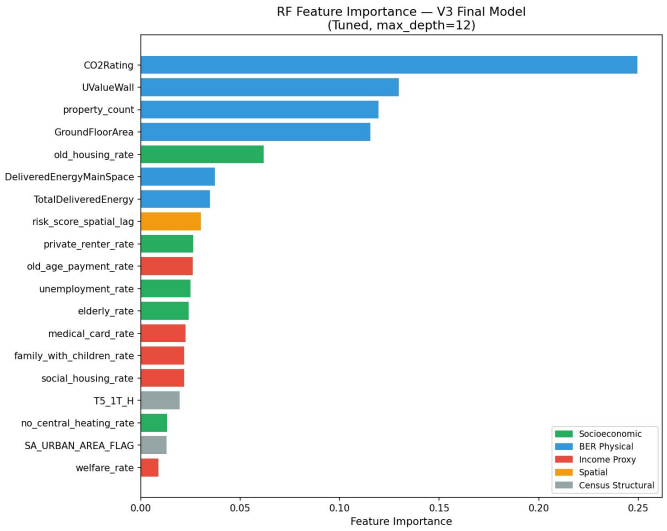

Fig. 5. RF feature importance for the final tuned model (`max_depth=12`). Features are coloured by category: BER Physical (blue), Socioeconomic (green), Income Proxy (red), Spatial (orange), Census Structural (grey).

the carbon intensity of the primary heating fuel, making it a compound indicator of physical and fuel vulnerability.

`property_count` (≈ 0.12) and GroundFloorArea (≈ 0.11) follow as BER physical features. `old_housing_rate` (≈ 0.06) is the highest-ranked socio-economic feature, consistent with dwelling vintage as the dominant driver of structural energy inefficiency [11]. The spatial lag `risk_score_spatial_lag` (≈ 0.03) provides meaningful additional signal, validating the inclusion of neighbourhood context [14]. Income proxy features each contribute in the 0.01–0.025 range, consistent with their role as indirect rather than direct indicators. The relatively low importance of `SA_URBAN_AREA_FLAG` (≈ 0.01) indicates that once CO_2 rating and wall U-value are controlled for, the urban–rural divide adds limited independent discriminatory power, building physics are more predictive than settlement geography per se.

C. Geographic Distribution

Fig. 6 presents the national energy poverty risk choropleth, classifying 4,026 Small Areas (21.3%) as High Risk. Rural western counties, particularly Donegal, Mayo, Galway, Clare, and Kerry, exhibit the densest clustering, consistent with the prevalence of pre-1950 stone-built housing with little or no insulation [16]. Significant high-risk clusters also appear

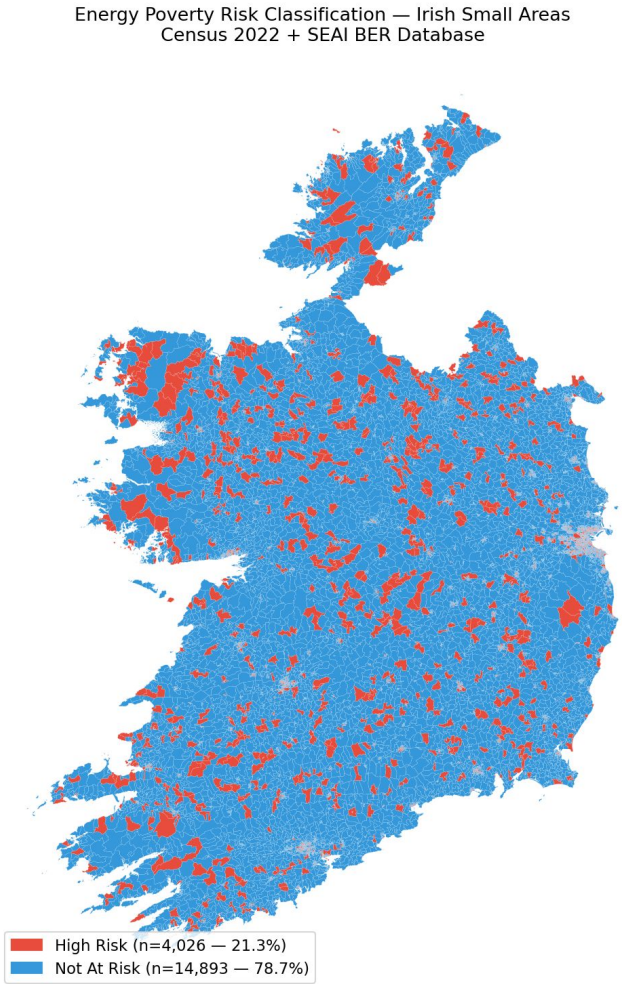

Fig. 6. Energy poverty risk classification across 18,916 Irish Small Areas. High Risk (red, $n = 4{,}026$, 21.3%) is concentrated in rural Western counties and inner-city Dublin; Not At Risk (blue, $n = 14{,}893$, 78.7%) dominates suburban and post-2000 development zones.

in inner-city Dublin, reflecting the juxtaposition of high-density older housing with concentrated deprivation, a pattern identified by Kelly et al. [12]. The Midlands (Laois, Offaly, Roscommon) show moderate but diffuse risk, reflecting a mix of rural housing vintage and moderate income levels.

The large Not At Risk zones correspond to post-2000 residential development in suburban Dublin, Cork, Limerick, and Galway, characterised by high BER ratings (B2 or above) built to modern Part L building regulations. This geographic face validity confirms that the classifier correctly identifies areas a domain expert would expect to be lower risk based on housing stock age and income level.

V. Conclusions and Future Work

This study developed a machine learning pipeline to classify energy poverty risk at Small Area level in Ireland using Building Energy Rating (BER) data and Census (SAPS 2022) socioeconomic indicators. The BER dataset was first extensively cleaned, including standardisation of SA_Code values, correction of multi-area entries, and recovery of valid records. The cleaned BER data were then aggregated to Small Area level and merged with Census variables to create a unified national dataset covering all 18,919 Small Areas.

Missing BER coverage (11.1% of Small Areas) was addressed using MissForest [18] imputation, improving dataset completeness and enabling full national-scale modelling. The final dataset was used to train a Random Forest classifier, which achieved a test accuracy of 77.40% and an AUC-ROC of 0.8393.

Feature analysis showed that physical building characteristics derived from BER data were the strongest predictors, followed by selected socioeconomic variables from Census data. This indicates that housing quality plays a central role in explaining variation in energy poverty risk across Small Areas, alongside broader socioeconomic conditions.

Overall, the study demonstrates that integrating BER and Census data at Small Area level produces a consistent and usable dataset for machine learning classification of energy poverty risk. However, limitations include the use of proxy variables rather than direct household income or energy consumption data, and remaining uncertainty introduced by imputation in areas with incomplete BER coverage. Future work could address these limitations by incorporating direct energy usage data and historical BER datasets to enable temporal analysis and improved validation of model predictions.

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